

Software Requirements Specification (SRS) - Blood Bank Management System (BBMS)

Project: Blood Bank Management System (BBMS) - Software Engineering Mini-Project, Deliverable Part 1

Version: 1.0

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Technology stack: Backend - Python (Django) or Node.js; Frontend - JavaScript / HTML / CSS; Database - SQLite

Revision history

Version	Date	Author	Change summary	Approval
1.0	27 Sep 2026	Project Team	Initial SRS for Deliverable Part 1	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, and security requirements for the Blood Bank Management System (BBMS), a web-based application developed to manage blood donation, testing, inventory, and hospital requisition workflows. This document is prepared as Deliverable Part 1 of a university Software Engineering mini-project and follows an IEEE Std 830-style structure.

1.2 Scope

BBMS is a standalone web application that supports the following in-scope capabilities:

- Donor registration, authentication, profile management, and appointment scheduling
- Medical eligibility screening prior to donation
- Blood collection recording, unique blood-unit (Bag) identification, and blood-group management
- Serology/infection testing, quarantine, and safe-disposal workflows
- Blood component separation and expiry management using First-Expiry-First-Out (FEFO)
- Real-time blood inventory tracking across blood bank locations
- Hospital registration, blood requisition (including emergency/STAT requests), compatibility cross-matching, allocation, dispatch, and delivery confirmation
- Role-based administration, audit logging, reporting/analytics, and public blood-availability lookup

The following are explicitly out of scope for this mini-project:

- Integration with national or regional blood-bank registries or government health systems
- Payment processing or billing of any kind
- Direct interfacing with physical laboratory testing equipment (results are entered manually by staff)
- Native mobile applications (the system is delivered as a responsive web application)
- Multi-language localization

1.3 Audience

This document is intended for the course instructor/evaluator, the student development team responsible for design and implementation, and any future maintainer of the BBMS codebase.

1.4 Definitions

Term	Definition
BBMS	Blood Bank Management System
SRS	Software Requirements Specification
FEFO	First-Expiry-First-Out
RBC	Red Blood Cells
RBAC	Role-Based Access Control
API	Application Programming Interface
REST	Representational State Transfer
HTTPS/TLS	Hypertext Transfer Protocol Secure / Transport Layer Security
ORM	Object-Relational Mapping
UI	User Interface
UML	Unified Modeling Language
XSS	Cross-Site Scripting
SQLi	SQL Injection
STAT	Immediate / emergency priority request

Term	Definition
Bag ID	Unique identifier assigned to a collected blood unit
WCAG	Web Content Accessibility Guidelines

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications — used as a structural reference for this document.
- General publicly available domain background on blood-bank operating workflows (donor screening, testing, FEFO inventory practice) — used only to inform realistic system behaviour, not as a certified regulatory source.

1.6 Document Overview

Section 2 describes the product at a high level. Section 3 defines external interfaces. Section 4 lists functional requirements by module. Section 5 lists non-functional requirements and, in Section 5.1, the security objectives and requirements. Section 6 covers quality attributes and acceptance tests. Section 7 presents UML use-case diagrams. Section 8 provides a requirements traceability matrix linking requirements to modules and test cases defined in the companion Test Plan.

2. Overall description

2.1 Product perspective

BBMS is a new, self-contained web application developed for academic demonstration purposes. It is not a replacement for, nor does it integrate with, any existing legacy blood-bank system. The system is architected to resemble a realistic blood-bank management workflow (donor – collection – testing – inventory – requisition – dispatch) at a scale appropriate for a university mini-project.

2.2 Major product functions

- Donor management (registration, profile, eligibility, appointments)
- Blood collection, testing, quarantine, and component separation
- Inventory management with expiry (FEFO) handling
- Hospital management and blood requisition/dispatch workflow
- Compatibility cross-matching for safe allocation
- Role-based administration and audit logging
- Reporting and analytics
- Public blood-availability lookup

2.3 User roles and characteristics

User Class	Description
Donor	A registered individual who books appointments, donates blood, and views personal donation history. Assumed to have basic computer literacy.

User Class	Description
Hospital User	A representative of a registered hospital who submits blood requisitions, tracks their status, and confirms delivery. Assumed to have basic computer literacy and domain familiarity with blood-component terminology.
Blood Bank Staff	Laboratory/collection personnel who record collections, perform and log serology testing, manage quarantine and component separation, manage inventory, approve requisitions, and perform allocation and dispatch.
Administrator	Manages user accounts and roles, approves hospital registrations, views the complete audit log, and generates system-wide reports.
Public / Guest	An unauthenticated visitor who may view the public blood-availability page only.

2.4 Operating environment

- Server: runs on a standard PC, laptop, or single virtual machine; no specialized hardware required.
- Backend runtime: Python 3.10+ (Django) or Node.js 18+ (Express), per the implementing team's technology choice.
- Database: SQLite (file-based, embedded), appropriate for the mini-project's scale.
- Client: any device with a modern, JavaScript-enabled web browser and internet/LAN connectivity to the server.

2.5 Constraints

- The system must be implemented using the specified stack: Python (Django) or Node.js backend, JavaScript/HTML/CSS frontend, SQLite database.
- SQLite has limited concurrent-write support; the system is designed for the concurrency levels of a university mini-project demonstration, not a production multi-hospital network.
- The project must be delivered by the fixed Deliverable Part 1 deadline of 28 September 2026, which constrains the depth of implementation (though not the completeness of this specification).
- The system is an academic prototype and is not certified for real clinical or regulatory use.

2.6 Assumptions and Dependencies

- Users have access to a device with internet or LAN connectivity and a modern web browser.
- Donors self-report accurate information during registration and the eligibility questionnaire.
- Physical laboratory testing (serology) is performed by blood bank staff using equipment outside the system; staff enter results into BBMS manually.
- All system components run with synchronized clocks (server time is authoritative for timestamps, expiry calculations, and audit logs).

3. External interfaces

3.1 User interfaces

BBMS provides a responsive, browser-based web interface with distinct dashboards for each user class (Donor, Hospital, Staff, Administrator). Navigation menus and available actions are rendered according to the authenticated user's role. All data-entry forms perform client-side validation in addition to server-side validation.

3.2 Hardware interfaces

BBMS does not interface directly with specialized hardware. Client devices require only standard input (keyboard/touch) and display capability sufficient to render a modern web browser. No barcode scanners, lab analyzers, or other devices are integrated in this mini-project scope.

3.3 Software interfaces

- Backend web framework: Django (Python) or Express (Node.js).
- Data access: an Object-Relational Mapping (ORM) layer (e.g., Django ORM or Sequelize/Prisma) over a SQLite database file.
- Outbound notification interface: SMTP for sending email notifications (registration confirmation, appointment updates, password reset, emergency-requisition alerts).

3.4 Communications

- All client-server communication shall use HTTPS.
- The frontend communicates with the backend via a RESTful JSON API (see the companion Software Architecture & Design Specification for endpoint details).
- Outbound email notifications are sent via SMTP.

4. System features (detailed)

Each functional requirement is uniquely identified, atomic, measurable, testable, and stated using "shall." Requirements are grouped by module below. Each requirement below includes acceptance criteria and a reference test case (defined in the companion Test Plan). IDs follow BBMS-F-###.

4.1 Donor Registration and Login

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-001	The system shall allow a new donor to register by providing full name, date of birth, gender, contact number, email address, and a password.	Functional	High	Donor	Verified by TC-AUTH-01, TC-AUTH-02 (expected results in STP)	—
BBMS-F-002	The system shall validate that the donor is at least 18 years old at the time of registration, calculated from the provided date of birth.	Functional	High	Donor	Verified by TC-AUTH-01, TC-AUTH-02 (expected results in STP)	—
BBMS-F-003	The system shall enforce a unique email address per donor account and shall reject registration attempts using an email already in use.	Functional	High	Donor	Verified by TC-AUTH-01, TC-AUTH-02 (expected results in STP)	—
BBMS-F-004	The system shall authenticate a registered donor using email and password before granting access to donor-only functions.	Functional	High	Donor	Verified by TC-AUTH-01, TC-AUTH-02 (expected results in STP)	—
BBMS-F-005	The system shall allow a donor to reset a forgotten password via a time-limited, single-use reset link sent to the registered email address.	Functional	Medium	Donor	Verified by TC-AUTH-01, TC-AUTH-02 (expected results in STP)	Requires SMTP email (SRS 3.3)

4.2 Donor Profile

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-006	The system shall allow a logged-in donor to view and update personal profile details, including contact number, address, and blood group.	Functional	Medium	Donor	Verified by TC-DNR-01 (expected results in STP)	—
BBMS-F-007	The system shall display the donor's donation history, including date, location, and Bag ID of each past donation.	Functional	Medium	Donor	Verified by TC-DNR-01 (expected results in STP)	—
BBMS-F-008	The system shall display the donor's next eligible donation date, calculated as a configurable minimum interval (default 90 days) after the last successful donation.	Functional	Medium	Donor	Verified by TC-DNR-01 (expected results in STP)	—

4.3 Medical Eligibility Questionnaire

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-009	The system shall present a standardized medical eligibility questionnaire to a donor before confirming an appointment booking.	Functional	High	Donor	Verified by TC-ELG-01 (expected results in STP)	—
BBMS-F-010	The system shall automatically mark a donor as temporarily ineligible if any questionnaire answer matches a configured disqualifying condition, and shall display the reason to the donor.	Functional	High	Donor	Verified by TC-ELG-01 (expected results in STP)	—
BBMS-F-011	The system shall record the questionnaire responses and eligibility outcome against the associated appointment for audit purposes.	Functional	Medium	Blood Bank Staff	Verified by TC-ELG-01 (expected results in STP)	—

4.4 Donation Appointment Booking, Rescheduling, and Cancellation

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-012	The system shall allow an eligible donor to book a donation appointment by selecting an available date, time slot, and blood bank location.	Functional	High	Donor	Verified by TC-APT-01, TC-APT-02 (expected results in STP)	Requires eligibility screening (F-009, F-010)
BBMS-F-013	The system shall prevent booking of a time slot that has reached its configured donor capacity.	Functional	High	Donor	Verified by TC-APT-01, TC-APT-02 (expected results in STP)	Slot capacity configured (F-012)
BBMS-F-014	The system shall allow a donor to reschedule an existing appointment to another available slot up to 24 hours before the scheduled time.	Functional	Medium	Donor	Verified by TC-APT-01, TC-APT-02 (expected results in STP)	Depends on F-012
BBMS-F-015	The system shall allow a donor to cancel an existing appointment and shall release the reserved slot capacity upon cancellation.	Functional	Medium	Donor	Verified by TC-APT-01, TC-APT-02 (expected results in STP)	Depends on F-012

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-016	The system shall send a confirmation notification (on-screen and/or email) when an appointment is booked, rescheduled, or cancelled.	Functional	Low	Donor	Verified by TC-APT-01, TC-APT-02 (expected results in STP)	SMTP email optional (SRS 3.3)

4.5 Blood Collection

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-017	The system shall allow blood bank staff to record the start and completion of a blood collection against a confirmed donor appointment.	Functional	High	Blood Bank Staff	Verified by TC-COL-01 (expected results in STP)	Requires confirmed appointment (F-012)
BBMS-F-018	The system shall require blood bank staff to record the collected volume (in millilitres) and collection date/time for each donation.	Functional	High	Blood Bank Staff	Verified by TC-COL-01 (expected results in STP)	—
BBMS-F-019	The system shall update the donor's last-donation date and donation history immediately upon confirming a successful collection.	Functional	High	Blood Bank Staff	Verified by TC-COL-01 (expected results in STP)	Depends on F-017, F-018

4.6 Unique Blood Unit / Bag ID

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-020	The system shall automatically generate a unique Bag ID for every collected blood unit at the time collection is recorded.	Functional	High	Blood Bank Staff	Verified by TC-BAG-01 (expected results in STP)	—
BBMS-F-021	The system shall prevent the creation of two blood units with the same Bag ID.	Functional	High	Blood Bank Staff	Verified by TC-BAG-01 (expected results in STP)	Depends on F-020

4.7 Blood Group Management

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-022	The system shall support the eight standard ABO/Rh blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-) for donor and blood-unit records.	Functional	High	Blood Bank Staff	Verified by TC-BAG-02 (expected results in STP)	—
BBMS-F-023	The system shall allow blood bank staff to record or correct the laboratory-confirmed blood group of a blood unit, distinct from the donor's self-reported blood group.	Functional	High	Blood Bank Staff	Verified by TC-BAG-02 (expected results in STP)	Distinct from self-reported group (F-006)

4.8 Serology / Infection Testing

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-024	The system shall require every collected blood unit to undergo recorded serology testing for HIV, Hepatitis B, Hepatitis C, and Syphilis before it can be marked available for use.	Functional	High	Blood Bank Staff	Verified by TC-SER-01, TC-SER-02 (expected results in STP)	—
BBMS-F-025	The system shall allow blood bank staff to record a Pass or Fail result for each serology test performed on a blood unit.	Functional	High	Blood Bank Staff	Verified by TC-SER-01, TC-SER-02 (expected results in STP)	—
BBMS-F-026	The system shall set the status of a blood unit to "Quarantined" by default upon collection and shall not permit a change to "Available" status until all required test results are recorded as Pass.	Functional	High	Blood Bank Staff	Verified by TC-SER-01, TC-SER-02 (expected results in STP)	Depends on F-024, F-025
BBMS-F-027	The system shall set the status of a blood unit to "Discarded" if any serology test result is recorded as Fail, and shall prevent such units from being allocated or dispatched.	Functional	High	Blood Bank Staff	Verified by TC-SER-01, TC-SER-02 (expected results in STP)	Depends on F-025

4.9 Quarantine of Unsafe Blood

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-028	The system shall exclude blood units with a "Quarantined" or "Discarded" status from inventory availability counts and hospital search results.	Functional	High	Blood Bank Staff	Verified by TC-QNT-01 (expected results in STP)	Depends on F-026, F-027
BBMS-F-029	The system shall log the identity of the staff member and timestamp whenever a blood unit's status changes to or from "Quarantined."	Functional	High	Blood Bank Staff	Verified by TC-QNT-01 (expected results in STP)	—
BBMS-F-030	The system shall allow authorized blood bank staff to view a list of all currently quarantined blood units together with the elapsed quarantine duration.	Functional	Medium	Blood Bank Staff	Verified by TC-QNT-01 (expected results in STP)	—

4.10 Blood Component Separation

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-031	The system shall allow blood bank staff to record the separation of a whole-blood unit into components (e.g., Red Blood Cells, Plasma, Platelets), generating a distinct inventory record for each resulting component.	Functional	Medium	Blood Bank Staff	Verified by TC-CMP-01 (expected results in STP)	—
BBMS-F-032	The system shall retain a reference from each separated component record to its originating whole-blood unit Bag ID for traceability.	Functional	Medium	Blood Bank Staff	Verified by TC-CMP-01 (expected results in STP)	Depends on F-031

4.11 Blood Inventory Management

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-033	The system shall maintain a real-time count of available blood units per blood group and component type at each blood bank location.	Functional	High	Blood Bank Staff	Verified by TC-INV-01, TC-INV-02 (expected results in STP)	—
BBMS-F-034	The system shall allow blood bank staff to view the complete inventory, filterable by blood group, component type, and status.	Functional	High	Blood Bank Staff	Verified by TC-INV-01, TC-INV-02 (expected results in STP)	—
BBMS-F-035	The system shall automatically decrement the available inventory count when a blood unit is allocated to a hospital requisition.	Functional	High	Blood Bank Staff	Verified by TC-INV-01, TC-INV-02 (expected results in STP)	Triggered by allocation (F-052)
BBMS-F-036	The system shall automatically increment the available inventory count when a new blood unit is marked "Available" after passing serology testing.	Functional	High	Blood Bank Staff	Verified by TC-INV-01, TC-INV-02 (expected results in STP)	Depends on F-026

4.12 Expiry Handling and FEFO

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-037	The system shall calculate and store an expiry date for each blood unit based on its component type at the time it is marked "Available" (e.g., 42 days for whole blood/RBC, 5 days for platelets, 1 year for frozen plasma).	Functional	High	Blood Bank Staff	Verified by TC-EXP-01 (expected results in STP)	Applied when unit becomes Available (F-026)
BBMS-F-038	The system shall apply First-Expiry-First-Out (FEFO) ordering when suggesting or automatically selecting blood units for allocation to a requisition.	Functional	High	Blood Bank Staff	Verified by TC-EXP-01 (expected results in STP)	Used by allocation (F-052)
BBMS-F-039	The system shall automatically flag a blood unit's status as "Expired" once its expiry date has passed and shall exclude expired units from availability counts and allocation.	Functional	High	Blood Bank Staff	Verified by TC-EXP-01 (expected results in STP)	—

4.13 Hospital Registration and Login

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-040	The system shall allow a hospital to register by providing hospital name, registration/license number, address, contact person, and contact details, subject to administrator approval before activation.	Functional	High	Hospital User	Verified by TC-AUTH-03 (expected results in STP)	—
BBMS-F-041	The system shall authenticate a registered hospital user using hospital-specific login credentials before granting access to requisition functions.	Functional	High	Hospital User	Verified by TC-AUTH-03 (expected results in STP)	—

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-042	The system shall prevent an unapproved hospital account from submitting blood requisitions.	Functional	High	Administrator	Verified by TC-AUTH-03 (expected results in STP)	Depends on F-040, F-062

4.14 Blood Requisition

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-043	The system shall allow an authenticated hospital user to submit a blood requisition specifying blood group, component type, quantity of units, and required-by date.	Functional	High	Hospital User	Verified by TC-REQ-01, TC-REQ-02 (expected results in STP)	—
BBMS-F-044	The system shall assign each submitted requisition a unique Requisition ID and an initial status of "Pending."	Functional	High	Hospital User	Verified by TC-REQ-01, TC-REQ-02 (expected results in STP)	—
BBMS-F-045	The system shall allow blood bank staff to approve, partially fulfil, or reject a pending requisition, recording a reason for any rejection.	Functional	High	Blood Bank Staff	Verified by TC-REQ-01, TC-REQ-02 (expected results in STP)	Depends on F-044
BBMS-F-046	The system shall allow a hospital user to view the current status and history of their submitted requisitions.	Functional	Medium	Hospital User	Verified by TC-REQ-01, TC-REQ-02 (expected results in STP)	—

4.15 Emergency / STAT Requests

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-047	The system shall allow a hospital user to mark a requisition as "Emergency/STAT," which shall place it at the top of the blood bank staff's pending-requisition queue.	Functional	High	Hospital User	Verified by TC-REQ-03 (expected results in STP)	—
BBMS-F-048	The system shall notify available blood bank staff, via on-screen alert and/or email, immediately upon submission of an Emergency/STAT requisition.	Functional	High	Blood Bank Staff	Verified by TC-REQ-03 (expected results in STP)	Depends on F-047; SMTP optional (SRS 3.3)

4.16 Blood Compatibility and Cross-Matching

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-049	The system shall determine and display ABO/Rh-compatible blood groups for a requested blood group before allocation, based on standard transfusion-compatibility rules.	Functional	High	Blood Bank Staff	Verified by TC-MATCH-01 (expected results in STP)	—

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-050	The system shall prevent blood bank staff from allocating a blood unit whose blood group is not compatible with the requisition's requested blood group, unless an authorized staff member records an explicit override with justification.	Functional	High	Blood Bank Staff	Verified by TC-MATCH-01 (expected results in STP)	Depends on F-049
BBMS-F-051	The system shall record the cross-matching outcome (Compatible / Incompatible / Overridden) against each allocation.	Functional	High	Blood Bank Staff	Verified by TC-MATCH-01 (expected results in STP)	Depends on F-050

4.17 Allocation and Dispatch

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-052	The system shall allow blood bank staff to allocate one or more specific blood units, by Bag ID, to an approved requisition.	Functional	High	Blood Bank Staff	Verified by TC-DISP-01 (expected results in STP)	Requires approved requisition (F-045)
BBMS-F-053	The system shall generate a dispatch record containing the requisition ID, allocated Bag ID(s), dispatching staff identity, and dispatch timestamp.	Functional	High	Blood Bank Staff	Verified by TC-DISP-01 (expected results in STP)	Depends on F-052
BBMS-F-054	The system shall update a dispatched blood unit's status to "Dispatched" and remove it from available inventory counts.	Functional	High	Blood Bank Staff	Verified by TC-DISP-01 (expected results in STP)	Depends on F-053
BBMS-F-055	The system shall update the requisition status to "Fulfilled" once all requested units have been dispatched, or to "Partially Fulfilled" if fewer units are dispatched than requested.	Functional	High	Blood Bank Staff	Verified by TC-DISP-01 (expected results in STP)	Depends on F-054

4.18 Delivery Confirmation

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-056	The system shall allow a hospital user to confirm receipt of a dispatched blood unit, recording the confirmation timestamp and receiving staff identity.	Functional	Medium	Hospital User	Verified by TC-DEL-01 (expected results in STP)	Depends on F-053
BBMS-F-057	The system shall update a blood unit's status to "Delivered" upon hospital confirmation of receipt.	Functional	Medium	Hospital User	Verified by TC-DEL-01 (expected results in STP)	Depends on F-056

4.19 Audit Logging

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-058	The system shall record an audit log entry for every creation, status change, or deletion of a donor, blood unit, requisition, or user account, capturing the acting user, action performed, timestamp, and affected record ID.	Functional	High	Administrator	Verified by TC-AUD-01 (expected results in STP)	—
BBMS-F-059	The system shall restrict access to the complete audit log to users with the Administrator role, and shall render audit log entries read-only (non-editable, non-deletable) to all users.	Functional	High	Administrator	Verified by TC-AUD-01 (expected results in STP)	Depends on F-058

4.20 Admin, User, and Role Management

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-060	The system shall allow an Administrator to create, view, deactivate, and reactivate user accounts for all roles (Donor, Hospital, Blood Bank Staff, Administrator).	Functional	High	Administrator	Verified by TC-ADM-01 (expected results in STP)	—
BBMS-F-061	The system shall enforce Role-Based Access Control (RBAC) such that each system function is accessible only to the user roles authorized for it, as defined in the system's role-permission matrix.	Functional	High	Administrator	Verified by TC-ADM-01 (expected results in STP)	—
BBMS-F-062	The system shall allow an Administrator to approve or reject pending hospital registration requests.	Functional	High	Administrator	Verified by TC-ADM-01 (expected results in STP)	Depends on F-040
BBMS-F-063	The system shall prevent a deactivated user account from authenticating into the system.	Functional	High	Administrator	Verified by TC-ADM-01 (expected results in STP)	Depends on F-060

4.21 Reports and Analytics

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-064	The system shall generate a summary report of current inventory levels by blood group and component type, exportable in CSV format.	Functional	Medium	Administrator	Verified by TC-RPT-01 (expected results in STP)	CSV export
BBMS-F-065	The system shall generate a donation activity report for a specified date range, showing total donations, units collected, and units discarded.	Functional	Medium	Administrator	Verified by TC-RPT-01 (expected results in STP)	—
BBMS-F-066	The system shall generate a requisition fulfilment report showing requisitions received, fulfilled, partially fulfilled, and rejected for a specified date range.	Functional	Medium	Administrator	Verified by TC-RPT-01 (expected results in STP)	—

4.22 Public Blood Availability

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BBMS-F-067	The system shall provide a public-facing page displaying current blood availability by blood group and blood bank location, accessible without authentication.	Functional	Medium	Public / Guest	Verified by TC-PUB-01 (expected results in STP)	—
BBMS-F-068	The system shall exclude donor-identifying and hospital-identifying information from the public blood availability page.	Functional	High	Donor / Hospital User (privacy)	Verified by TC-PUB-01 (expected results in STP)	Applies to F-067

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to the test plan. IDs follow BBMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement	Comments / Dependencies
BBMS-NF-001	The system shall return results for a blood-availability search within 2 seconds under normal load (up to 50 concurrent users).	Performance	High	Measured by TC-NF-01 (method and threshold in STP)	—
BBMS-NF-002	The system shall complete a donor appointment-booking transaction, from request to confirmation, within 3 seconds.	Performance	Medium	Measured by TC-NF-01 (method and threshold in STP)	—
BBMS-NF-003	The system shall support at least 50 concurrent authenticated users without response times exceeding the limits stated in BBMS-NF-001.	Performance	Medium	Measured by TC-NF-01 (method and threshold in STP)	Uses limits of BBMS-NF-001
BBMS-NF-004	The system shall be available for use at least 99% of scheduled operating hours, excluding planned maintenance windows.	Reliability / Availability	Medium	Measured by TC-NF-02 (method and threshold in STP)	—
BBMS-NF-005	The system shall retain all committed transaction data (donations, requisitions, dispatches) across an application restart, with no data loss for committed transactions.	Reliability / Availability	High	Measured by TC-NF-02 (method and threshold in STP)	—
BBMS-NF-006	The system shall log unhandled application errors, with timestamp and stack trace, to a server-side error log for diagnosis.	Reliability / Availability	Medium	Measured by TC-NF-02 (method and threshold in STP)	—
BBMS-NF-007	A first-time user shall be able to complete donor registration without external assistance in under 5 minutes, as verified by a usability walkthrough.	Usability / Accessibility	Medium	Measured by TC-NF-03 (method and threshold in STP)	—
BBMS-NF-008	The system's web interface shall conform to WCAG 2.1 Level AA colour-contrast and keyboard-navigation guidelines for all primary user workflows.	Usability / Accessibility	Medium	Measured by TC-NF-03 (method and threshold in STP)	—
BBMS-NF-009	The system shall display clear, human-readable error messages for all form-validation failures, identifying the specific field and issue.	Usability / Accessibility	High	Measured by TC-NF-03 (method and threshold in STP)	—
BBMS-NF-010	The system's backend shall follow a layered/modular code structure such that a change to one module (e.g., inventory) does not require modification of unrelated modules (e.g., reporting).	Maintainability	Medium	Measured by TC-NF-04 (method and threshold in STP)	—
BBMS-NF-011	The system shall maintain a minimum of 60% automated unit-test coverage for backend business-logic modules.	Maintainability	Low	Measured by TC-NF-04 (method and threshold in STP)	—

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement	Comments / Dependencies
BBMS-NF-012	The system's web client shall render correctly on the latest two stable versions of Chrome, Firefox, and Edge.	Portability	Medium	Measured by TC-NF-05 (method and threshold in STP)	—
BBMS-NF-013	The system's backend shall be deployable on any host operating system supporting Python 3.10+ or Node.js 18+ without code modification.	Portability	Low	Measured by TC-NF-05 (method and threshold in STP)	—
BBMS-NF-014	The system's database schema shall support migration from SQLite to a multi-user RDBMS (e.g., PostgreSQL) without change to the application's data-access interfaces.	Scalability	Low	Measured by TC-NF-06 (method and threshold in STP)	—
BBMS-NF-015	The system's architecture shall allow the addition of a new blood bank location without modification of the existing donor, hospital, or inventory modules.	Scalability	Low	Measured by TC-NF-06 (method and threshold in STP)	—

5.1 Security

5.1.1 Security Objectives

- Protect the confidentiality of donor personal and medical information from unauthorized access or disclosure.
- Ensure that only authenticated and authorized users can perform actions appropriate to their assigned role.
- Preserve the integrity of blood-unit status and testing records to prevent unsafe blood from being allocated or dispatched.
- Maintain accountability through a non-repudiable audit trail of all security-relevant and data-changing actions.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
BBMS-SR-001	The system shall require all users (Donor, Hospital, Blood Bank Staff, Administrator) to authenticate with a unique email/username and password before accessing any role-specific function.	Security	High	Covers: Authentication. Verified by TC-SEC-01 (expected results in STP)
BBMS-SR-002	The system shall store user passwords using a salted, one-way hashing algorithm (e.g., bcrypt) and shall never store or transmit passwords in plaintext.	Security	High	Covers: Password hashing. Verified by TC-SEC-02 (expected results in STP)
BBMS-SR-003	The system shall enforce role-based access control (RBAC) on the server side for every API endpoint, independent of client-side UI restrictions, and shall deny unauthorized requests with an HTTP 403 response.	Security	High	Covers: RBAC / authorization. Verified by TC-SEC-03 (expected results in STP)
BBMS-SR-004	The system shall use parameterized queries or an Object-Relational Mapping (ORM) layer for all database operations and shall not construct SQL statements by concatenating raw user input.	Security	High	Covers: SQL injection prevention. Verified by TC-SEC-04 (expected results in STP)

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
BBMS-SR-005	The system shall validate and sanitize all user-supplied input on both client and server sides, and shall encode output rendered in the browser to prevent stored and reflected Cross-Site Scripting (XSS).	Security	High	Covers: Input validation / XSS. Verified by TC-SEC-05 (expected results in STP)
BBMS-SR-006	The system shall issue session tokens as HttpOnly, Secure cookies with a defined expiry, and shall invalidate a session upon logout or after 30 minutes of inactivity.	Security	High	Covers: Session / cookie security. Verified by TC-SEC-06 (expected results in STP)
BBMS-SR-007	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts and shall rate-limit login requests to no more than 10 per minute per source IP address.	Security	High	Covers: Brute-force / rate limiting. Verified by TC-SEC-07 (expected results in STP)
BBMS-SR-008	The system shall transmit all client-server communication over HTTPS/TLS and shall record a tamper-evident audit trail (per BBMS-F-058) of security-relevant events, including login attempts, account lockouts, and role changes.	Security	High	Covers: Confidentiality, integrity, audit. Verified by TC-SEC-08, TC-AUD-01 (expected results in STP)

6. Quality attributes & Acceptance tests

Quality attributes (Section 5): Performance (NF-001–NF-003); Reliability and Availability (NF-004–NF-006); Usability and Accessibility (NF-007–NF-009); Maintainability (NF-010–NF-011); Portability (NF-012–NF-013); Scalability (NF-014–NF-015); Security (SR-001–SR-008).

Exit criteria for acceptance (per the companion Test Plan):

- All planned functional and security test cases have been executed or formally marked Blocked with reason.
- No unresolved Critical or High severity defect remains for an acceptance-critical feature.
- All mandatory performance thresholds in NF-001 through NF-003 have been evaluated.
- Reliability, usability/accessibility, portability, maintainability, and scalability requirements have been assessed using their defined methods.
- The RTM (Section 8) is updated with actual test status and defect references.

Acceptance test suites:

Suite	Test cases
Functional (by module)	TC-AUTH-01/02/03, TC-DNR-01, TC-ELG-01, TC-APT-01/02, TC-COL-01, TC-BAG-01/02, TC-SER-01/02, TC-QNT-01, TC-CMP-01, TC-INV-01/02, TC-EXP-01, TC-REQ-01/02/03, TC-MATCH-01, TC-DISP-01, TC-DEL-01, TC-AUD-01, TC-ADM-01, TC-RPT-01, TC-PUB-01
Security	TC-SEC-01 to TC-SEC-08
Performance	TC-NF-01
Reliability / Availability	TC-NF-02
Usability / Accessibility	TC-NF-03
Maintainability	TC-NF-04
Portability	TC-NF-05
Scalability	TC-NF-06

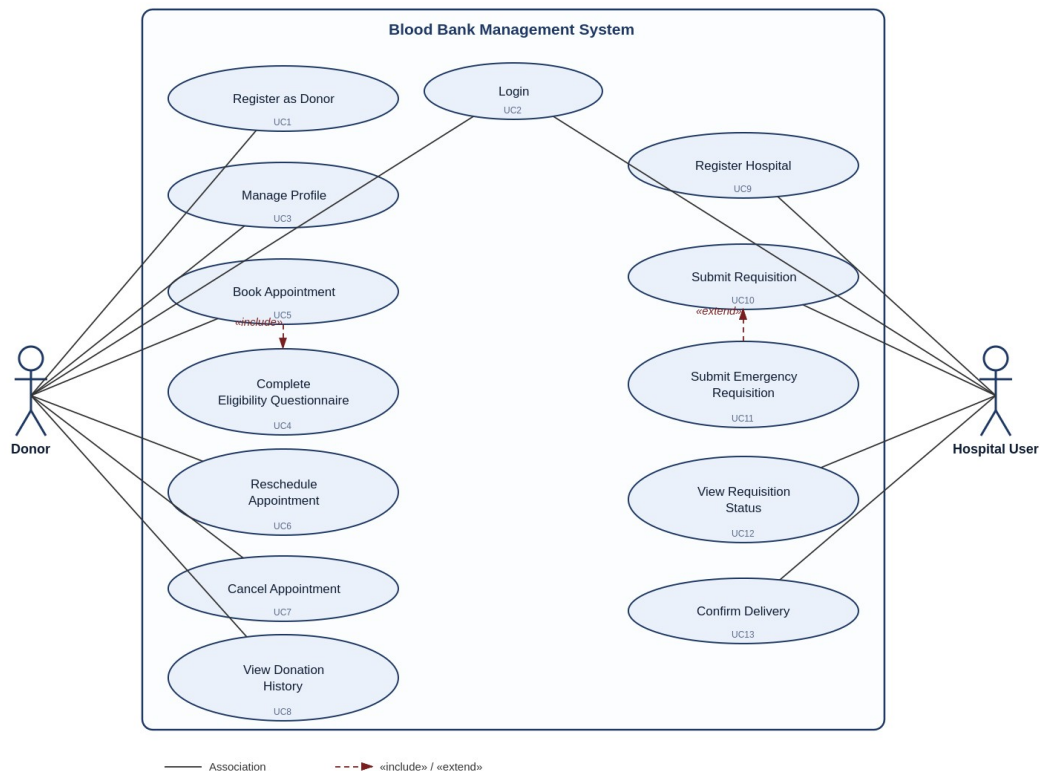
7. System models and diagrams

The system boundary is the Blood Bank Management System (BBMS). Equivalent Mermaid source is included after each diagram so it can also be pasted into a Markdown/GitHub viewer.

7.1 UML Use-Case diagram

7.1 Use-Case Diagram — Donor and Hospital Workflows

Actors: Donor, Hospital User. “Book Appointment” includes “Complete Eligibility Questionnaire” (BBMS-F-009). “Submit Emergency Requisition” extends “Submit Requisition” (BBMS-F-047).



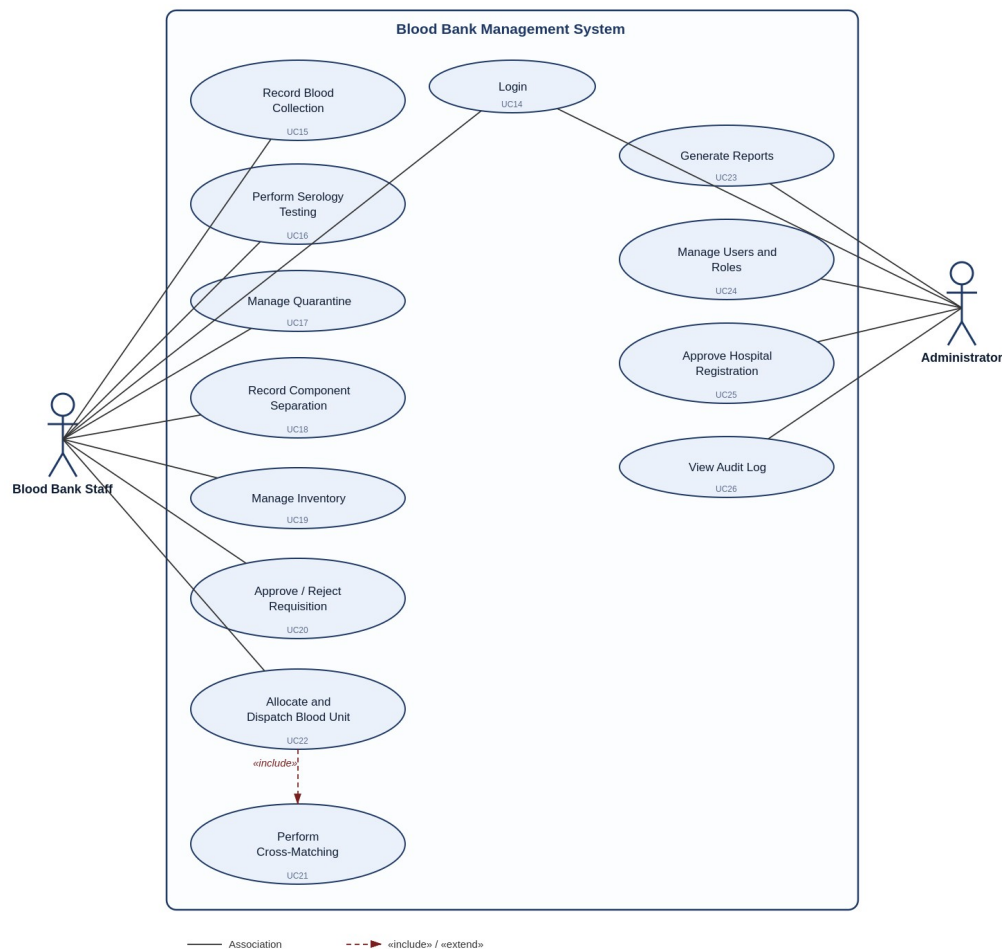
Mermaid source (7.1)

```
flowchart LR
    Donor([Donor])
    Hospital([Hospital User])
    subgraph BBMS[Blood Bank Management System]
        UC1([UC1: Register as Donor])
        UC2([UC2: Login])
        UC3([UC3: Manage Profile])
        UC4([UC4: Complete Eligibility Questionnaire])
        UC5([UC5: Book Appointment])
        UC6([UC6: Reschedule Appointment])
        UC7([UC7: Cancel Appointment])
        UC8([UC8: View Donation History])
        UC9([UC9: Register Hospital])
        UC10([UC10: Submit Requisition])
        UC11([UC11: Submit Emergency Requisition])
        UC12([UC12: View Requisition Status])
        UC13([UC13: Confirm Delivery])
    end
    Donor --- UC1
    Donor --- UC3
    Donor --- UC5
    Donor --- UC6
    Donor --- UC7
    Donor --- UC8
    Hospital --- UC9
    Hospital --- UC10
    Hospital --- UC11
    Hospital --- UC12
    Hospital --- UC13
    UC2 --- Donor
    UC2 --- Hospital
    UC5 -.->|«include»| UC4
    UC11 -.->|«extend»| UC10
end
```


Donor --> UC1
 Donor --> UC2
 Donor --> UC3
 Donor --> UC5
 Donor --> UC6
 Donor --> UC7
 Donor --> UC8
 UC5 -.include.-> UC4
 Hospital --> UC9
 Hospital --> UC2
 Hospital --> UC10
 Hospital --> UC12
 Hospital --> UC13
 UC11 -.extend.-> UC10

7.2 Use-Case Diagram — Staff and Administrator Workflows

Actors: Blood Bank Staff, Administrator. “Allocate and Dispatch Blood Unit” includes “Perform Cross-Matching” (BBMS-F-049).



Mermaid source (7.2)

```

    flowchart LR
      Staff([Blood Bank Staff])
      Admin([Administrator])
      subgraph BBMS [Blood Bank Management System]
        UC14(Login)
        UC15(Record Blood Collection)
        UC16(Perform Serology Testing)
        UC17(Manage Quarantine)
        UC18(Record Component Separation)
        UC19(Manage Inventory)
        UC20(Approve / Reject Requisition)
        UC22(Allocate and Dispatch Blood Unit)
        UC21(Perform Cross-Matching)
        UC23(Generate Reports)
        UC24(Manage Users and Roles)
        UC25(Approve Hospital Registration)
        UC26(View Audit Log)
      end
      Staff --- UC15
      Staff --- UC16
      Staff --- UC17
      Staff --- UC18
      Staff --- UC19
      Staff --- UC20
      Staff --- UC22
      Staff --- UC21
      Admin --- UC14
      Admin --- UC23
      Admin --- UC24
      Admin --- UC25
      Admin --- UC26
      UC22 -.-> UC21 : «include»
  
```

```

UC15(Record Blood Collection)
UC16(Perform Serology Testing)
UC17(Manage Quarantine)
UC18(Record Component Separation)
UC19(Manage Inventory)
UC20(Approve / Reject Requisition)
UC21(Perform Cross-Matching)
UC22(Allocate and Dispatch Blood Unit)
UC23(Generate Reports)
UC24(Manage Users and Roles)
UC25(Approve Hospital Registration)
UC26(View Audit Log)
end
Staff --> UC14
Staff --> UC15
Staff --> UC16
Staff --> UC17
Staff --> UC18
Staff --> UC19
Staff --> UC20
Staff --> UC22
UC22 -.include.-> UC21
Admin --> UC14
Admin --> UC23
Admin --> UC24
Admin --> UC25
Admin --> UC26

```

8. Requirements Traceability Matrix (RTM)

This matrix maps each requirement group to its module and to the Test Case ID(s) defined in the companion Test Plan. Status values follow the Test Plan's convention ("Not Run", "Pass", "Fail", "Blocked") and are set to "N" (Not Run) pending execution.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
BBMS-F-001-F-005	Donor Registration / Login	4.1 / SAD 3.4	Authentication & RBAC	TC-AUTH-01, TC-AUTH-02	N (Not Run)	
BBMS-F-006-F-008	Donor Profile	4.2 / SAD 3.4	Donor Management	TC-DNR-01	N (Not Run)	
BBMS-F-009-F-011	Medical Eligibility Questionnaire	4.3 / SAD 3.4	Donor Management	TC-ELG-01	N (Not Run)	
BBMS-F-012-F-016	Appointment Booking / Reschedule / Cancel	4.4 / SAD 3.4	Donor Management	TC-APT-01, TC-APT-02	N (Not Run)	
BBMS-F-017-F-019	Blood Collection	4.5 / SAD 3.4	Donor Management	TC-COL-01	N (Not Run)	
BBMS-F-020-F-021	Unique Blood Unit / Bag ID	4.6 / SAD 3.4	Inventory / FEFO	TC-BAG-01	N (Not Run)	
BBMS-F-022-F-023	Blood Group Management	4.7 / SAD 3.4	Inventory / FEFO	TC-BAG-02	N (Not Run)	
BBMS-F-024-F-027	Serology / Infection Testing	4.8 / SAD 3.4	Testing & Quarantine	TC-SER-01, TC-SER-02	N (Not Run)	
BBMS-F-028-F-030	Quarantine of Unsafe Blood	4.9 / SAD 3.4	Testing & Quarantine	TC-QNT-01	N (Not Run)	
BBMS-F-031-F-032	Blood Component Separation	4.10 / SAD 3.4	Testing & Quarantine	TC-CMP-01	N (Not Run)	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
BBMS-F-033-F-036	Blood Inventory Management	4.11 / SAD 3.4	Inventory / FEFO	TC-INV-01, TC-INV-02	N (Not Run)	
BBMS-F-037-F-039	Expiry Handling / FEFO	4.12 / SAD 3.4	Inventory / FEFO	TC-EXP-01	N (Not Run)	
BBMS-F-040-F-042	Hospital Registration / Login	4.13 / SAD 3.4	Hospital & Requisition	TC-AUTH-03	N (Not Run)	
BBMS-F-043-F-046	Blood Requisition	4.14 / SAD 3.4	Hospital & Requisition	TC-REQ-01, TC-REQ-02	N (Not Run)	
BBMS-F-047-F-048	Emergency / STAT Requests	4.15 / SAD 3.4	Hospital & Requisition	TC-REQ-03	N (Not Run)	
BBMS-F-049-F-051	Compatibility / Cross-Matching	4.16 / SAD 3.4	Compatibility & Allocation	TC-MATCH-01	N (Not Run)	
BBMS-F-052-F-055	Allocation and Dispatch	4.17 / SAD 3.4	Compatibility & Allocation; Dispatch & Delivery	TC-DISP-01	N (Not Run)	
BBMS-F-056-F-057	Delivery Confirmation	4.18 / SAD 3.4	Dispatch & Delivery	TC-DEL-01	N (Not Run)	
BBMS-F-058-F-059	Audit Logging	4.19 / SAD 3.4	Audit & Reporting	TC-AUD-01	N (Not Run)	
BBMS-F-060-F-063	Admin / User / Role Management	4.20 / SAD 3.4	Authentication & RBAC	TC-ADM-01	N (Not Run)	
BBMS-F-064-F-066	Reports and Analytics	4.21 / SAD 3.4	Audit & Reporting	TC-RPT-01	N (Not Run)	
BBMS-F-067-F-068	Public Blood Availability	4.22 / SAD 3.4	Public Availability	TC-PUB-01	N (Not Run)	
BBMS-NF-001-NF-003	Performance	5 / SAD 2.5	Web UI / REST API / SQLite	TC-NF-01	N (Not Run)	
BBMS-NF-004-NF-006	Reliability / Availability	5 / SAD 2.5	Application host / SQLite	TC-NF-02	N (Not Run)	
BBMS-NF-007-NF-009	Usability / Accessibility	5 / SAD 2.5	Web Client	TC-NF-03	N (Not Run)	
BBMS-NF-010-NF-011	Maintainability	5 / SAD 2.5	All backend modules	TC-NF-04	N (Not Run)	
BBMS-NF-012-NF-013	Portability	5 / SAD 2.5	Web Client / Application host	TC-NF-05	N (Not Run)	
BBMS-NF-014-NF-015	Scalability	5 / SAD 2.5	ORM / Data Access; Inventory	TC-NF-06	N (Not Run)	
BBMS-SR-001	Authentication	5.1.2 / SAD 3	Authentication & RBAC	TC-SEC-01	N (Not Run)	
BBMS-SR-002	Password Hashing	5.1.2 / SAD 3	Authentication & RBAC	TC-SEC-02	N (Not Run)	
BBMS-SR-003	RBAC / Authorization	5.1.2 / SAD 3	Authentication & RBAC / REST API	TC-SEC-03	N (Not Run)	
BBMS-SR-004	SQL Injection Prevention	5.1.2 / SAD 3	ORM / Data Access	TC-SEC-04	N (Not Run)	
BBMS-SR-005	Input Validation / XSS	5.1.2 / SAD 3	REST API / Web Client	TC-SEC-05	N (Not Run)	
BBMS-SR-006	Session / Cookie Security	5.1.2 / SAD 3	Authentication & RBAC	TC-SEC-06	N (Not Run)	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
BBMS-SR-007	Brute-Force / Rate Limiting	5.1.2 / SAD 3	Authentication & RBAC	TC-SEC-07	N (Not Run)	
BBMS-SR-008	Audit / Confidentiality / Integrity	5.1.2 / SAD 3	Audit & Reporting / TLS	TC-SEC-08, TC-AUD-01	N (Not Run)	